

Rino Genistus Ruban

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EDUCATION

Rutgers University

New Brunswick, NJ

B.S. Computer Science, Minor in Mathematics

Expected Jan 2027

- Relevant coursework: Machine Learning Principles, Data Science, Computational Robotics, Design and Analysis of Algorithms

PROJECTS

Medical RAG LLM for Study Assistance | *React, FastAPI, Pinecone, Python*

2026

- Engineered a hybrid Retrieval-Augmented Generation pipeline integrating BM25 keyword matching with vector embeddings to deliver high-precision semantic search across 20,000+ processed text chunks from multiple medical reference textbooks.
- Built a custom document ingestion framework with recursive chunking and automated data sanitization to efficiently process dense academic texts at scale.
- Designed a full-stack React + FastAPI application integrated with Pinecone vector storage, enabling real-time, source-grounded responses with dynamic performance and UI.

Document Vector Search Engine | *Python*

2026

- Developed a local search engine in Python that indexes and retrieves text documents using Vector Space Modeling and Cosine Similarity, including custom weighting heuristic that prioritizes file names.
- Implemented a text processing pipeline including tokenization, stop-word removal, and regular expression-based cleaning to normalize data for high-accuracy matching.

Breast Cancer Classification — Multi-Model Pipeline | *Python, scikit-learn*

2026

- Built a comparative ML pipeline evaluating Logistic Regression (with feature scaling), Random Forest (with and without hyperparameter tuning via GridSearchCV), and Gradient Boosting on the Wisconsin Breast Cancer dataset.
- Applied StandardScaler preprocessing and cross-validation; analyzed precision/recall tradeoffs given the high cost of false negatives in medical diagnosis.

Deepfake Detection via Transfer Learning | *PyTorch, OpenCV, scikit-learn*

2026

- Authored a comprehensive technical report evaluating transfer learning across ResNet18, EfficientNet-B4, and ViT-Base architectures on the FaceForensics++ dataset, achieving a peak 97.43% classification accuracy.
- Built preprocessing and augmentation pipelines including OpenCV frame extraction, stratified dataset splitting, inverse-frequency class weighting, and random erasing.
- Utilized GradCAM visualizations across architectures to interpret model attention patterns and isolate facial regions driving the network's classification decisions.

Neural Network from Scratch | *Python*

2026

- Implemented a fully connected neural network in pure Python — 2-input, 2-neuron hidden layer, 1-output — with manual forward pass, loss computation, and backpropagation.
- Reinforced foundational understanding of gradient descent and weight updates without ML framework abstractions.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS, C++, C, Java

ML / AI: scikit-learn, NumPy, Pandas, PyTorch, CNN Architectures, ResNet

Web / Deploy: React, Vite, Netlify, REST APIs, FastAPI

LLM / RAG / Agentic AI: Transformer architecture, Pinecone, Vector DBs, Ollama, Multi-Agent systems, Tool-Use Routing

Tools: Git, GitHub, Jupyter, VS Code

ACTIVITIES

Rutgers AI Club

2025 – Present

Member

Rutgers Formula Racing

2024 – Present

Aerodynamics Team, Member